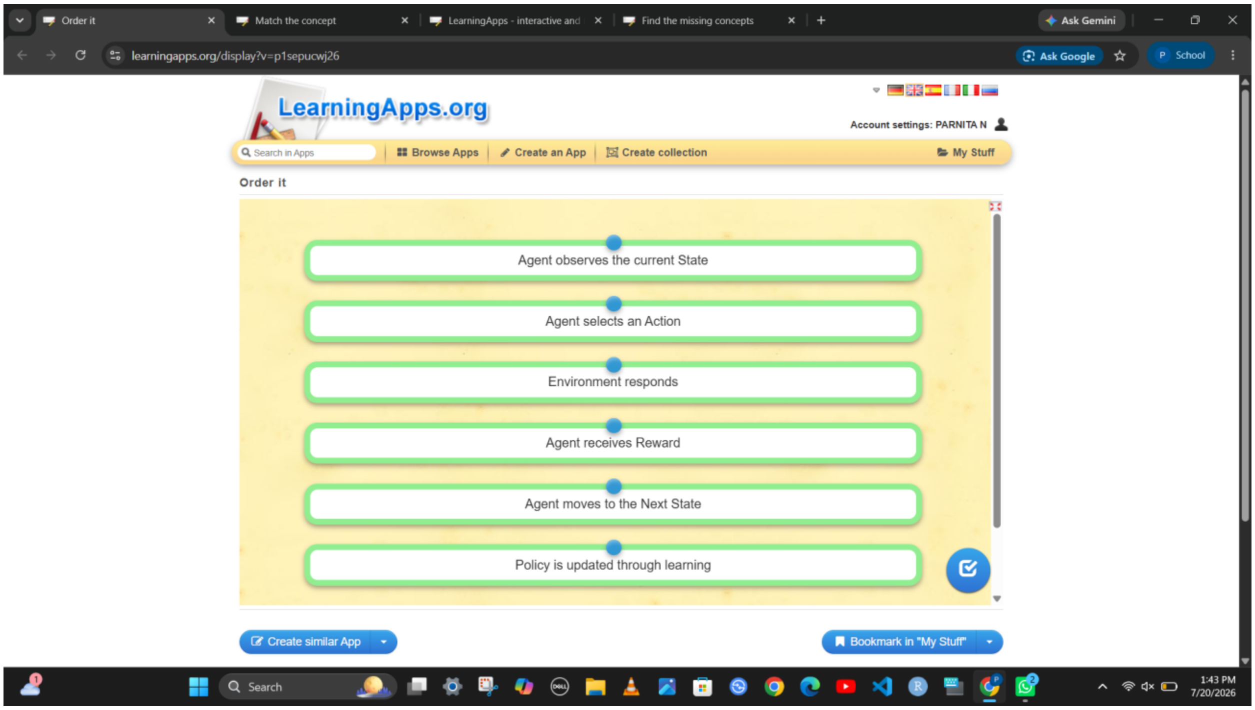
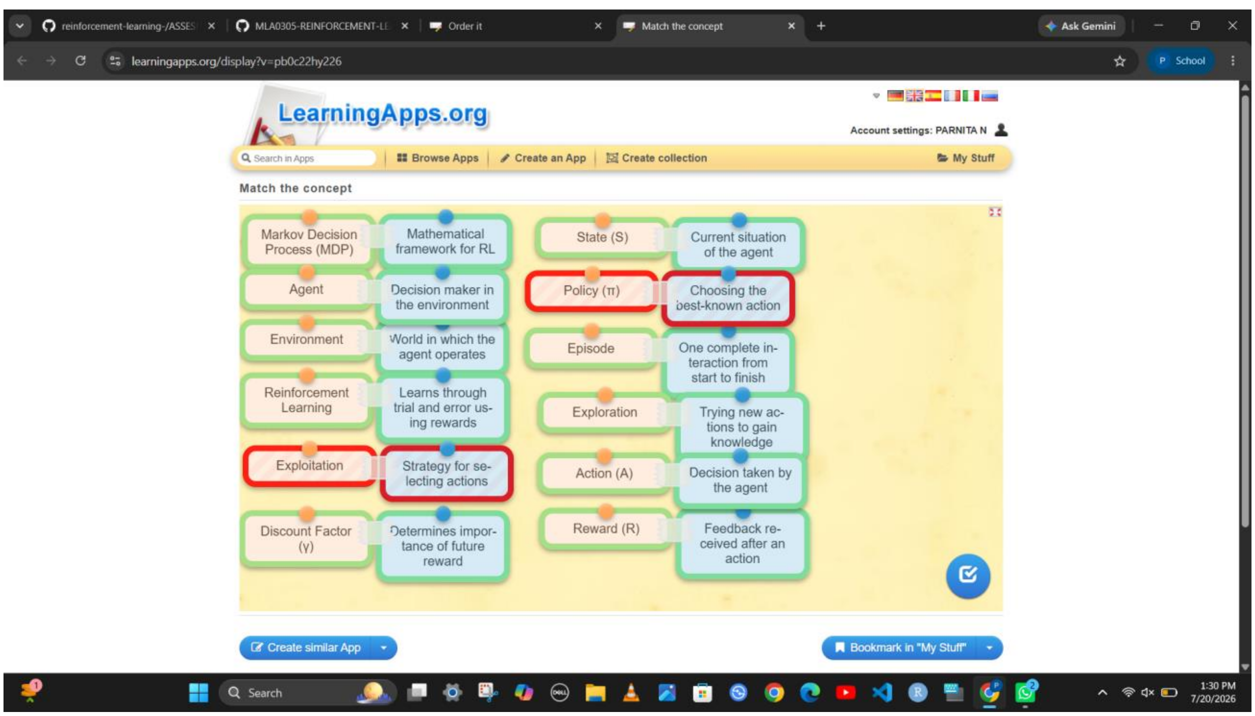
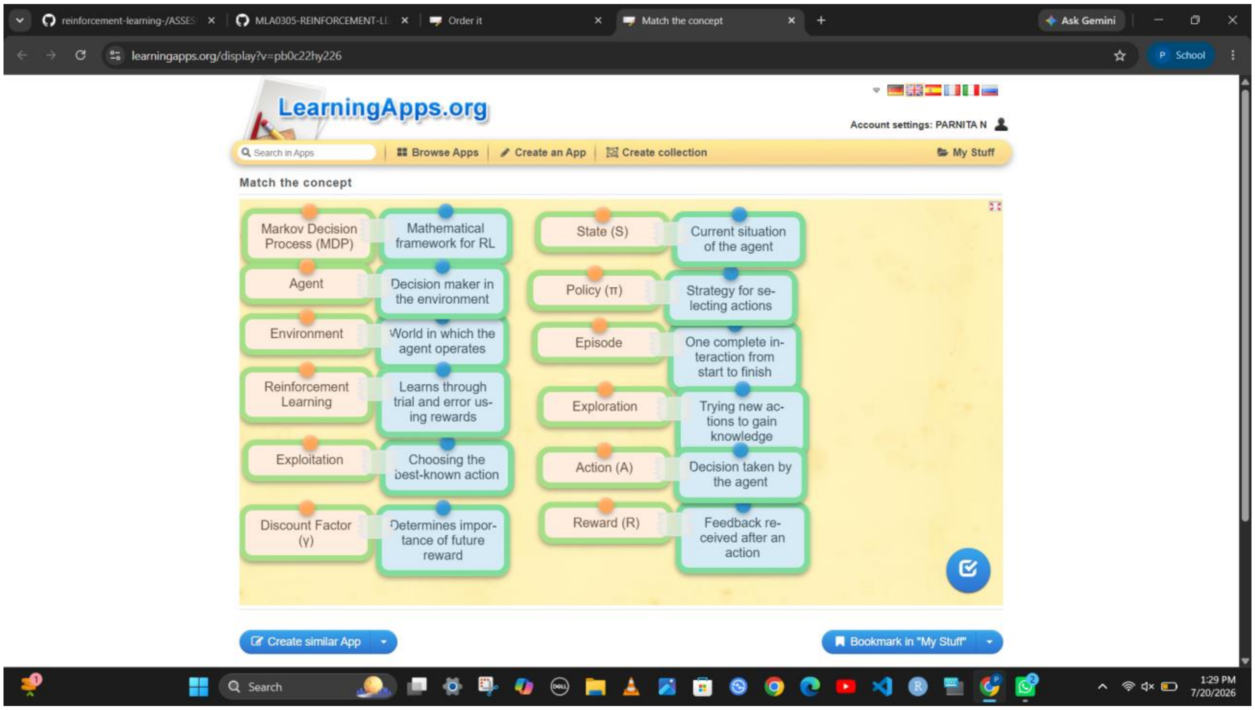
CO1 - AT1 - Part 2 Class Activities



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Markov Decision Process (MDP)
Mathematical framework for RL
Agent
Decision maker in the environment
Environment
World in which the agent operates
Reinforcement Learning
Learns through trial and error using rewards
Exploitation
Choosing the best-known action
Discount Factor (γ)
Determines importance of future reward
State (S)
Current situation of the agent
Policy (π)
Strategy for selecting actions
Episode
One complete interaction from start to finish
Exploration
Trying new actions to gain knowledge
Action (A)
Decision taken by the agent
Reward (R)
Feedback received after an action
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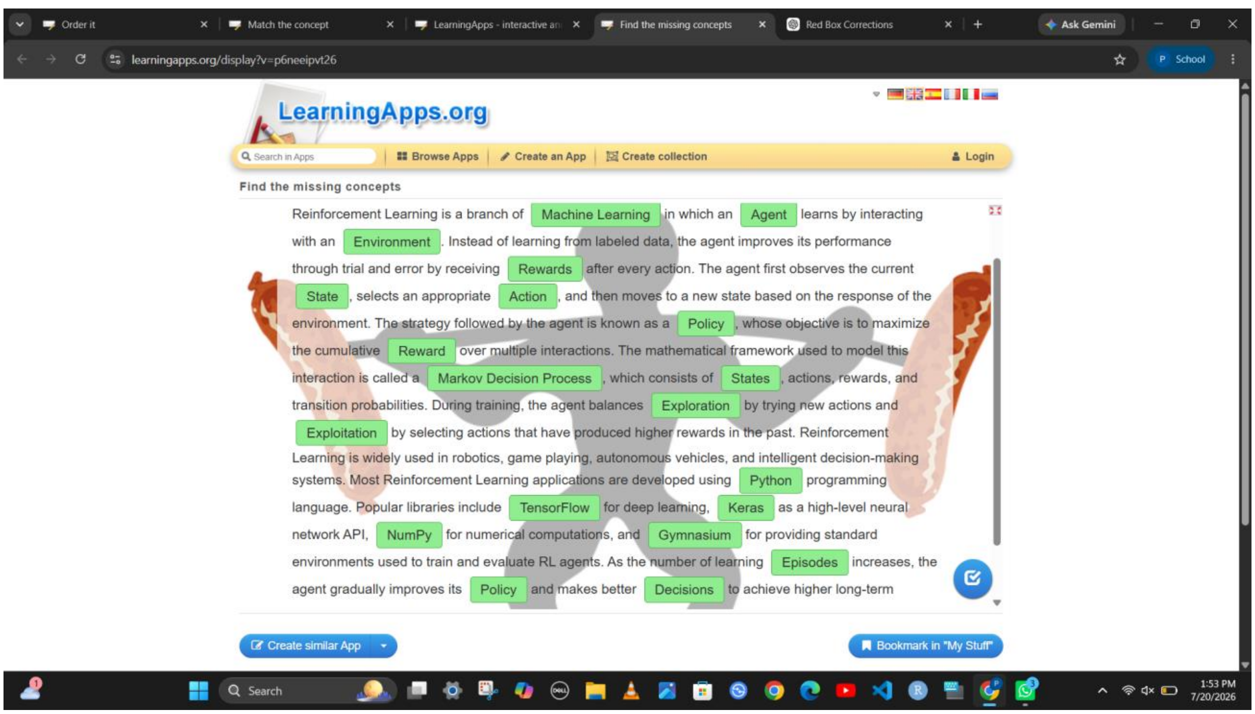
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Reinforcement Learning is a branch of Machine Learning in which an Agent learns by interacting with an Environment. Instead of learning from labeled data, the agent improves its performance through trial and error by receiving Rewards after every action. The agent first observes the current State, selects an appropriate Action, and then moves to a new state based on the response of the environment. The strategy followed by the agent is known as a Policy, whose objective is to maximize the cumulative Reward over multiple interactions. The mathematical framework used to model this interaction is called a Markov Decision Process, which consists of States, actions, rewards, and transition probabilities. During training, the agent balances Exploration by trying new actions and Exploitation by selecting actions that have produced higher rewards in the past. Reinforcement Learning is widely used in robotics, game playing, autonomous vehicles, and intelligent decision-making systems. Most Reinforcement Learning applications are developed using Python programming language. Popular libraries include TensorFlow for deep learning, Keras as a high-level neural network API, NumPy for numerical computations, and Gymnasium for providing standard environments used to train and evaluate RL agents. As the number of learning Episodes increases, the agent gradually improves its Policy and makes better Decisions to achieve higher long-term
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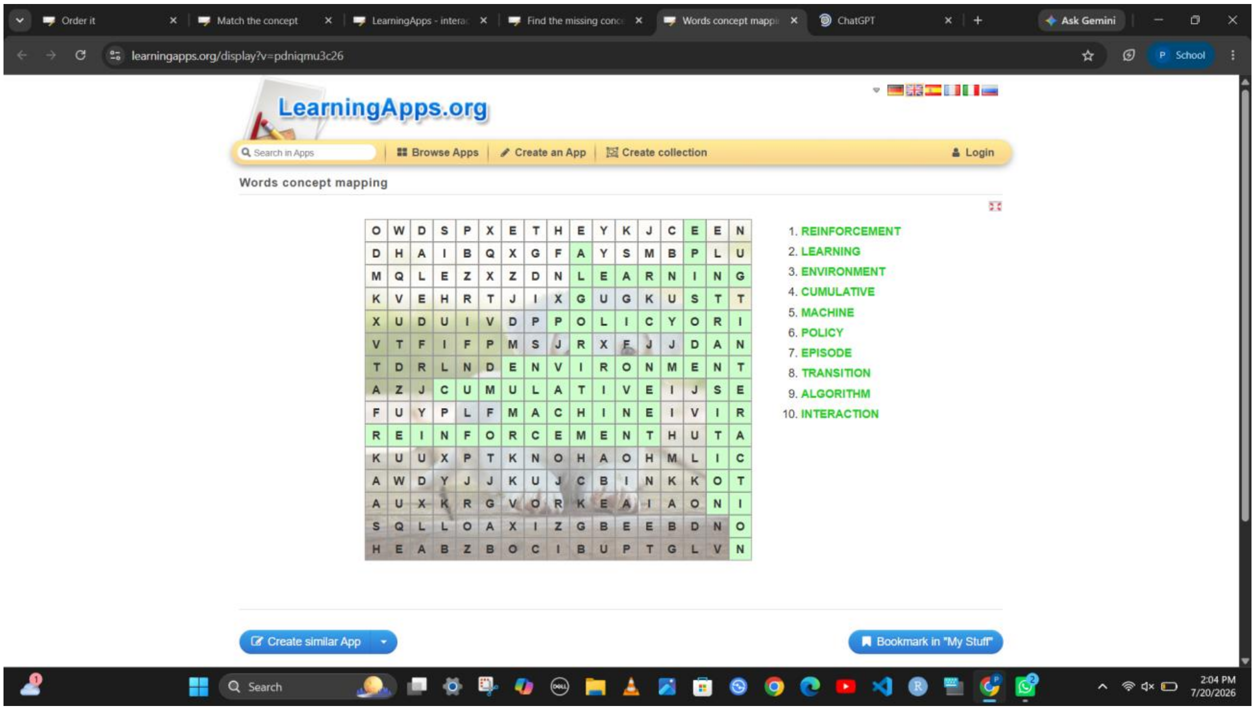
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1. REINFORCEMENT
2. LEARNING
3. ENVIRONMENT
4. CUMULATIVE
5. MACHINE
6. POLICY
7. EPISODE
8. TRANSITION
9. ALGORITHM
10. INTERACTION
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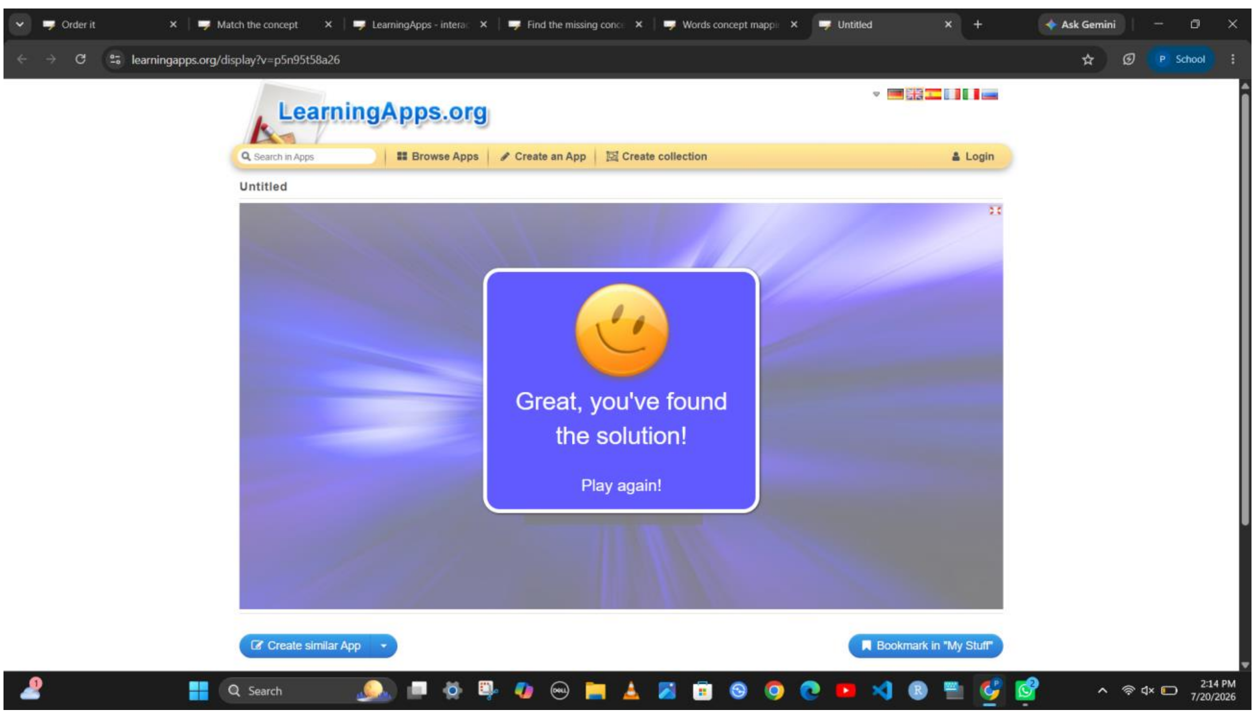
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